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Effects of storage temperature on silage nutrient composition, fermentation profile, and aerobic stability through a meta-analysis

G. F. L. Cruz,[1,2] M. R. Pupo,[1] J. L. P. Daniel,[3] and L. F. Ferraretto[1]*
[1]Department of Animal and Dairy Sciences, University of Wisconsin, Madison, WI 53706
[2]Animal Science Department, Federal University of Viçosa, Viçosa, MG, Brazil 36570-900
[3]Department of Animal Science, State University of Maringá, Maringá, PR, Brazil 87020-900

ABSTRACT

Silage is a major component of dairy diets worldwide, but the effects of storage temperature (ST) on the nutrient composition, fermentation profile, and aerobic stability of silage remain unclear. Our objectives were to (1) evaluate the effects of storage temperature on nutritive value and fermentation profile of silage, and (2) assess the effects of ST on nutritive value, fermentation profile, and aerobic stability of whole-plant corn silage (WPCS). Data from 42 peer-reviewed articles met the selection criteria, with a total of 253 treatment means. For WPCS, a secondary meta-analysis with data from 15 articles met the selection criteria, with a total of 96 treatment means. Silage was classified by type, storage length, and temperature range. Storage temperatures were categorized into 10°C increments: (1) ≤10°C; (2) >10 and ≤20°C; (3) >20 and ≤30°C; (4) >30 and ≤40°C; and (5) >40°C. Data were analyzed using a mixed-effects model with ST as a fixed effect and treatment (other than ST) within study as a random effect. Both storage length and silage type were fitted as continuous and categorical covariates in the model, respectively. For the WPCS subset, silage type classification was not included in the model. The pooled SEM for response variables per study was included as a weighting factor for the data and calculated as the inverse of the square of the variance. Means were determined using the LSMEANS statement in SAS 9.4 and were compared using the sequentially rejective Bonferroni *t*-test adjustment. Increasing ST impaired silage fermentation by reducing lactic acid bacteria (LAB) counts. Conversely, lower temperatures (≤10°C) restricted silage fermentation due to lower acid production and altered metabolic activity of LAB. For WPCS, increasing ST gradually increased DM losses, and greater ethanol production was observed for silage stored at 11°C to 20°C than at 31°C to 40°C. Overall, moderate temperatures (21°C to 30°C) may be more suitable for silage storage due to greater acid production, rapid pH decline, and lower DM losses. In addition, changes in fermentation profile across ST ranges indicate a potential shift in dominant bacterial species during ensiling. These findings can provide useful information about silage production for dairy farmers. In cold climates, producers should allow longer storage times to ensure adequate fermentation, and the use of inoculants capable of stimulating fermentation at low temperatures may provide additional benefits. Conversely, in warm or tropical regions, rapid silo sealing and careful face management are important to prevent excessive heating and nutrient losses during feed-out.

Key words: ensiling, lactic acid bacteria, silage nutritive value, whole-plant corn

INTRODUCTION

Silage is a major component of dairy diets worldwide. Several strategies have been pursued to improve silage nutritive value and minimize nutrient losses during storage (Wilkinson et al., 2003). However, environmental conditions before ensiling can influence both DM yield and nutritive value of crops harvested for silage production (Hornick, 1992). Moreover, these effects may extend into the ensiling process, thereby affecting silage fermentation profile (Bernardes et al., 2018). Previous studies have reported that storage temperature (**ST**) can have a greater effect on the microbiology of silage compared with microbial inoculants or the density of the ensiled material (Bai et al., 2022a,b). Lactic acid bacteria (**LAB**) are the most abundant microorganisms found in well-preserved silage and capable of surviving across a wide temperature spectrum. Despite the ability of LAB to grow under a wide temperature range (5–45°C; Oude Elferink et al., 2001), a recent study by Kim et al. (2021) reported that most LAB exhibit optimal growth between 25 and 40°C. In addition, other studies have reported that species can be highly active around 5°C (Duru et al., 2021) or above 45°C (Zheng et al., 2020). In addition to

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*Corresponding author: ferraretto@wisc.edu

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microbial activity, temperature influences plant-derived proteases that remain active during the early stages of ensiling. At moderate temperatures, these enzymes accelerate proteolysis, thereby reducing substrate availability for LAB (Tao et al., 2012). Conversely, high temperatures can denature enzymes and shift microbial metabolism toward less desirable fermentations (Robinson, 2015). The magnitude of these effects also depends on forage species, as grasses and legumes differ in buffering capacity, soluble carbohydrate concentration, and intrinsic enzyme profiles, which together interact with temperature to regulate pH decline and fermentation dynamics (Weinberg et al., 2001; Kim and Adesogan, 2006).

Generally, producers are often challenged by an increase in silage temperature at the start of ensiling due to continuous plant respiration and the activity of aerobic microorganisms when oxygen is still present in the plant material (Bernardes et al., 2018). This occurs more frequently when crops are ensiled in tropical regions (Adesogan, 2009) or under hot conditions in temperate climates (Kung, 2011). High environmental temperatures increase respiration and epiphytic microbial metabolism, which in turn generates heat before anaerobiosis is established (McDonald et al., 1991). Inadequate packing or delayed sealing can further exacerbate these temperature rises (Kung, 2011). Silage storage at high temperatures may reduce lactic acid formation and aerobic stability and increase pH and DM loss (Weinberg et al., 2001; Ashbell et al., 2002). Moreover, lower average daily temperature can also negatively affect the fermentation process. Such silage commonly has an impaired acid production (Kung, 2010), slower pH decline (Zhou et al., 2016), and high yeast counts (Weinberg and Muck, 1996). Although environmental temperature cannot be controlled from harvesting through storage, understanding the magnitude of response due to changes in temperature can provide useful information about silage production for dairy farmers. Such knowledge can support the implementation of management strategies to reduce these nutritive losses under challenging conditions (Ashbell et al., 2002) or guide future developments of technologies to alleviate these issues.

Even though reviews are available in the literature evaluating the effects of field and pre-ensiling conditions on silage production and fermentation patterns (Bernardes et al., 2018; Borreani et al., 2018), little is known about the effects of ST on the nutrient composition, fermentation profile, and aerobic stability of silage. Therefore, the primary objective of this study was to evaluate the effects of ST on nutritive value and fermentation profile of different types of silage. But given that whole-plant corn silage (**WPCS**) is the predominant forage used in dairy diets in the United States and other areas worldwide (FAO, 2024; USDA-NASS, 2025), our second objective was to assess the effects of storage temperature on nutritive value, fermentation profile, and aerobic stability of WPCS. Our hypothesis was that both low (≤10°C) and high (>40°C) ST ranges would influence nutritive value, fermentation profile, and aerobic stability of silage regardless of forage type.

MATERIALS AND METHODS

A comprehensive search of the literature published in the English language from 1984 to 2024 was conducted to identify studies evaluating silage management under different storage temperatures. A literature search was conducted using US National Library of Medicine and National Institutes of Health through PubMed (http://www.ncbi.nlm.nih.gov/pubmed), ISI Web of Science (http://apps.webofknowledge.com), and Google Scholar (https://scholar.google.com/) online databases with the keywords "silage temperature," "silage fermentation temperature," and "silage storage temperature." Furthermore, references reported in the collected manuscripts were also checked. In addition, further clarification of the data was obtained through personal communications with the authors. Results from the online search and personal communications were assessed individually for the initial screening to be considered for the meta-analysis.

Inclusion Criteria

For inclusion in the database, the studies were required to include (1) temperature evaluation within fixed or small range temperatures; (2) temperature remained constant during the storage period; (3) tested simultaneously at least 2 storage temperatures as treatment means; and (4) evaluated at least 12 h of ensiling period to ensure changes in silage fermentation. Then, a review of the materials and methods indicated the studies in which effects of treatments on silage fermentation were not reported, and these studies were not further included in the analysis.

Data Extraction

One hundred and five peer-reviewed papers, including several experiments, were assessed for eligibility, and 62 publications were excluded due to (1) treatments with large variation of daily temperatures (8 experiments), (2) inconsistent temperatures when the assigned storage length was reached (4 experiments), (3) ST not being considered treatment means (34 experiments), and (4) publications not being available in peer-reviewed journals (16 experiments). The flow of data collection for the meta-analysis is presented in Figure 1.

Data from 42 peer-reviewed manuscripts met the selection criteria, with a total of 253 treatment means used

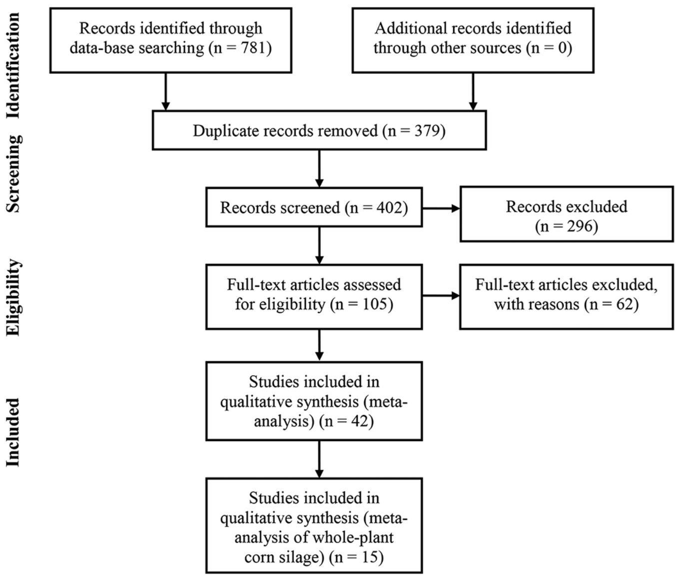

Figure 1. Preferred reporting items for systematic reviews and meta-analysis (PRISMA) flow diagram adapted from Moher et al. (2009). The flow describes the systematic review from initial search and screening to final selection of publications to be included in the meta-analysis. The 42 articles selected for inclusion in the meta-analysis contained 1 or multiple experiments.

to assess the effects of ST on nutritive value and fermentation profile of silage (Table 1). The experiment with WPCS conducted by Weinberg et al. (2001) contained a treatment with large temperature fluctuations throughout the storage period, and that treatment was therefore excluded from the dataset. Similarly, a treatment including oscillating ST from Zhou et al. (2019) was not included. The total number of treatment means (n) was corrected accordingly for both peer-reviewed manuscripts. Silage was further classified by type (whole-plant corn, grass, cereal, legume, TMR silage, and others [crop residues, byproducts, specific plant materials]), storage length (continuous variable ranging from 1 to 141 d), and temperature range. Temperature ranges were grouped in 10°C increases as follows: (1) ≤10°C; (2) >10 and ≤20°C; (3) >20 and ≤30°C; (4) >30 and ≤40°C; and (5) >40°C.

An additional meta-analysis was performed using a subset of studies that evaluated WPCS as the silage source. Data from 15 peer-reviewed manuscripts met the selection criteria, with a total of 96 treatment means used to evaluate the effects of ST on silage nutritive value, fermentation profile, and aerobic stability of WPCS. Classifications for both storage length and temperature range were conducted as previously described.

The number of replicates, treatment means, and sources of errors were extracted for the following response variables: pH, DM loss, concentrations of DM, NDF, ADF, CP, soluble CP (**SCP**), water-soluble carbohydrates (**WSC**), ash, lignin, lactic acid, acetic acid, lactic-to-acetic acid ratio, propionic acid, n-butyric acid, and ethanol, counts of LAB, yeast and mold (log cfu/g fresh matter), and aerobic stability (h). Articles with reported NH_3-N concentration on a DM basis were converted to % N. Similarly, total N concentrations were converted to % CP (CP = N × 6.25). The DM recovery (% DM) was converted to DM loss [DM loss (% DM) = 100 − DM recovery]. The OM (% DM) was converted to ash concentration [Ash (% DM) = 100 – OM]. Data transferred to Excel spreadsheets (Microsoft Excel 2016, Microsoft Corp., Redmond, WA) were reviewed by 2 trained people to ensure information collected was accurately transcribed from the manuscripts into the spreadsheets before statistical analysis. Data were screened for outliers based on visual assessment of variable distributions and evaluation of means and SD of variables expected to be normally distributed (Bateki et al., 2020).

Statistical Analysis

All the articles in the dataset were checked to determine whether SEM, SD, SE of the difference, or root mean square error (**RMSE**) were reported. In the 4 articles (1 with WPCS as silage source) that reported SD, SEM was calculated as described by Livingston (2004):

$$\mathrm{SEM} = \frac{\mathrm{SD}}{\sqrt{n}},$$

where n is the number of replications per treatment.

Data were analyzed using the MIXED procedure of SAS (version 9.4; SAS Institute Inc., Cary, NC) with the unstructured covariance structure, which provided the best fit according to Sawa's Bayesian information criterion. The mixed model included ST range as fixed effect and treatment (other treatments not including ST) within study as a random effect. A random intercept was specified for each study to account for between-study variability, allowing each study to have its own baseline level of the response variable, thereby adjusting for the hierarchical structure of the data (Pustejovsky et al., 2014). Both storage length and silage type (whole-plant corn, grass, cereal, legume, TMR silage, and others, such as crop residues, byproducts, and specific plant materials) were fitted as continuous and categorical covariates in the model, respectively. Covariates were checked for multicollinearity through Pearson correlation (de Souza et al., 2018).

The statistical model was represented as

$$Y_{ijk} = \mu + I_i + \beta^1 X^1{}_{ijk} + \beta^2 X^2{}_{ijk} + S_j + b_{jk} + \varepsilon_{ijk},$$

where Y_{ijk} is the observed response from the kth treatment within the jth study and ith temperature interval; μ is the

Table 1. Dataset used for meta-analysis of effects of storage temperature on silage nutrient composition, fermentation profile, and aerobic stability

Reference	Study	Experiment	Forage type[1]	Ensiling period (d)	Temperature (°C)
Ali et al. (2015)	1	1	WPC	28, 56, 112	5, 12, 18
		2			
Bai et al. (2022a)	2	3	WPC	60	20, 30
Bai et al. (2022b)	3	4	WPC	60	10, 25
Bolsen et al. (1988)	4	5	Alfalfa	90	15.5, 32.2
		6			
		7			
Chai et al. (2022)	5	8	Oat	60	25, 35, 45
Colombatto et al. (2004)	6	9	WPC	120	17.5, 40
Coskuntuna et al. (2010)	7	10	Wet brewers grain	45	20, 30, 37
Cruz et al. (2025)	8	11	WPC	7, 15, 30, 90	22, 40
Ferrero et al. (2021)	9	12	WPC	15, 30, 100	20, 14.6, 12.2, 4.6
Guan et al. (2020)	10	13	WPC	3, 7, 14, 70	30, 45
Gulfam et al. (2017)	11	14	Napier grass	50	30, 40, 50
Hou and Nishino (2022)	12	15	Guinea grass	3, 30, 60	25, 40
		16			
Hu et al. (2024)	13	17	Sorghum	2, 5, 15, 40	10, 20, 30, 40
Kim and Adesogan (2006)	14	18	WPC	82	20, 40
Koc et al. (2009)	15	19	WPC	45	20, 30, 37
		20	Vetch-grain	45	
Kondo et al. (2016)	16	21	TMR	30, 90	15, 30
Li et al. (2019a)	18	23	King grass, paspalum, white popinac, and stylo	30, 60	28, 40
Li et al. (2019b)	17	22	Corn stalk	7, 14, 30, 60	25, 40
Li et al. (2021)	19	24	Oat	60	5, 10, 15, 25
Liu et al. (2011)	20	25	Stylo	45	10, 20, 30, 40
Liu et al. (2012)	21	26	Stylo	45	20, 30, 40
Liu et al. (2016)	22	27	Napier grass	70	15, 30, 45
Liu et al. (2018)	23	28	Alfalfa	1, 3, 7, 21, 39, 65	15, 30, 45
Liu et al. (2024)	24	29	Alfalfa	60	20, 35
Muck and Dickerson (1988)	25	30	Alfalfa	40	15, 25, 35
		31			
		32			
		33			
Ren et al. (2020a)	27	35	Cauliflower leaf	30	20, 25, 30, 35, 40, 45
Ren et al. (2020b)	26	34	Maize straw + cabbage waste	30, 60, 90, 120	−3, 18, 34
Tamada et al. (1999)	28	36	Napier grass	45	40, 30
		37			
		38			
		39			
		40			
Tian et al. (2023)	29	41	TMR	56	15, 30, 40
Wang and Nishino (2013)	30	42	TMR	10, 30, 90	5, 15, 25, 35
Wang et al. (2018a)	32	44	WPC	55	20, 28, 37
Wang et al. (2018b)	31	43	Italian ryegrass	30	10, 15, 25
Wang et al. (2019)	33	45	Moringa	60	15, 30
Weinberg et al. (1998)	34	46	Wheat	60	25, 41
Weinberg et al. (2001)	35	47	WPC	63	28, 37
		48	Wheat	60	24, 41
Weiss et al. (2016)	36	49	WPC	141	20, 35
Zhang et al. (2010)	37	50	Barley straw	50	20, 30
		51			
		52	Guinea grass		
		53			
Zhang et al. (2018)	38	54	Alfalfa	60	20, 30, 40
Zhou et al. (2016)	39	55	WPC	60	5, 10, 15, 20, 25
	40	56		28	
Zhou et al. (2019)	41	57	WPC	60	10, 20
Zhu et al. (2022)	42	58	Oat	60	5, 15

[1]WPC: whole-plant corn.

overall mean; I_i is the fixed effect of storage temperature interval; $\beta^1X^1_{ijk}$ and $\beta^2X^2_{ijk}$ represent the fixed covariate effects of storage length (continuous) and silage type (categorical), respectively; S_j is the random effect of study; b_{jk} is the random effect of treatment nested within study; and ε_{ijk} is the residual error term.

For the WPCS subset, silage type classification was not included in the model. The pooled SEM for response variables per study was included as a weighting factor for the data and calculated as the inverse of the square of the variance (Roman-Garcia et al., 2016). Degrees of freedom were calculated using the Kenward–Roger option. The heterogeneity obtained from mixed-effect models for each outcome variable was calculated using a Q-test and represents underlying differences among studies around biological responses that are not explained by random error. Therefore, the degree of total variability for the outcome variables was evaluated with the I^2 statistic (Higgins and Thompson, 2002). Briefly, the I^2 statistics are calculated by transforming the square root of the Q-test divided by its df. Publication bias was assessed statistically as proposed by Egger et al. (1997). Means were determined using the LSMEANS statement in SAS and were compared using the sequentially rejective Bonferroni *t*-test adjustment after an overall significant F-test. Significance was considered at $P \leq 0.05$. Residual plots used data adjusted for the random effect of study, and the linear regressions were weighted for SEM to check for patterns. Accuracy was evaluated based on the RMSE, which was calculated by the plotted adjusted observation against the actual treatment observations (St. Pierre, 2001).

RESULTS

Data from 42 peer-reviewed manuscripts were collected to assess the effects of ST on silage nutritive value, fermentation profile, and aerobic stability of whole-plant corn (27.6% of the studies), whole-plant sorghum (1.7%), other cereals (8.6%), grasses (22.4%), alfalfa (17.2%), other legumes (8.6%), TMR (5.2%), and byproducts/other silage (8.6%). A total of 28.6% of the studies evaluated the effects of temperature at different storage lengths, and 66.6% of studies had storage lengths greater than 60 d. Most of the studies had ST varying from 21°C to 30°C (78.6%) and 11°C to 20°C (71.4%).

Silage type and storage length were not correlated ($P > 0.10$) and were retained in the mixed-effects model. Descriptive statistics for continuous variables included in the analysis are presented in Tables 2 and 3. A considerable variation in nutritive value, fermentation profile, microbial counts, and aerobic stability of silage reflects underlying differences in crop management, environmental conditions, silage inoculation, differences in experimental design and analytical methods, and statistical variation around variables of interest in the studies used in the current analysis.

The effect of ST was associated with large heterogeneity from the mixed-effect models. More than 90% of the total variability of the ST effect in nutritive value (DM, CP, SCP, NDF, and starch) across several forage types was due to heterogeneity ($I^2 > 90\%$). This indicates differences underlying the results of the studies used in the current analysis. Heterogeneity reflects underlying differences in environmental conditions, silage management, experimental design and analytical methods, and

Table 2. Descriptive statistics of silage nutritive value and fermentation profile of treatment means used in the meta-analysis

Item[1]	N[2]	Mean	SD	Minimum	Maximum
DM, % as-fed	412	32.1	11.3	12.5	61.2
CP, % DM	214	9.95	5.17	2.64	27.5
SCP, % DM	38	48.3	12.2	22.5	68.1
NDF, % DM	256	46.2	15.2	3.87	75.1
ADF, % DM	258	29.1	11.0	2.73	53.6
Lignin, % DM	90	4.22	2.26	1.34	14.4
Starch, % DM	74	35.3	8.26	8.21	44.9
WSC, % DM	225	3.03	3.13	0.09	14.0
Ash, % DM	98	4.10	1.52	2.30	8.70
DM loss, % DM	162	5.19	4.42	0.43	23.3
pH	548	4.44	0.70	3.41	7.07
Lactic acid, % DM	568	4.43	2.99	0.00	18.5
Acetic acid, % DM	568	1.50	1.48	0.00	10.8
LA:AA ratio	238	3.27	1.93	0.01	10.6
Propionic acid, % DM	214	0.18	0.28	0.00	2.43
n-Butyric acid, % DM	182	0.63	0.92	0.00	3.89
Ethanol, % DM	162	0.87	0.71	0.01	5.54
1,2-Propanediol, % DM	46	0.04	0.07	0.00	0.33
NH_3-N, % N	448	7.50	6.42	0.16	45.6
LAB, log cfu/g	274	6.77	1.85	0.00	10.9
Yeast, log cfu/g	224	3.13	2.37	0.00	7.68
Mold, log cfu/g	133	1.82	1.73	0.00	5.80

[1]SCP = soluble CP; WSC = water-soluble carbohydrates; LA:AA = lactic-to-acetic acid ratio; NH_3-N = ammonia-nitrogen; LAB = lactic acid bacteria.

[2]Number of treatment means.

Table 3. Descriptive statistics of whole-plant corn silage nutritive value, fermentation profile, and aerobic stability of treatment means used in the meta-analysis

Item[1]	N[2]	Mean	SD	Minimum	Maximum
DM, % as-fed	125	35.2	5.67	21.8	43.3
CP, % DM	96	6.88	0.82	9.10	5.60
SCP, % DM	22	47.0	14.3	22.5	68.1
NDF, % DM	94	40.7	7.78	30.8	59.0
ADF, % DM	94	21.3	4.49	16.4	37.8
Lignin, % DM	22	5.06	0.48	4.31	6.15
Starch, % DM	68	36.0	8.23	8.21	44.9
WSC, % DM	80	3.66	3.62	0.40	14.1
Ash, % DM	90	3.83	1.21	2.30	7.52
DM loss, % DM	70	3.95	2.84	1.20	13.3
pH	177	4.28	0.75	3.41	6.64
Lactic acid, % DM	175	4.90	2.10	0.83	10.2
Acetic acid, % DM	175	1.79	1.58	0.00	8.00
LA:AA ratio	143	3.85	2.02	0.91	10.6
Propionic acid, % DM	63	0.16	0.23	0.00	0.83
Ethanol, % DM	68	1.04	0.38	0.20	1.97
1,2-Propanediol, % DM	46	0.04	0.07	0.00	0.33
NH_3-N, % N	129	5.07	3.72	0.16	19.7
LAB, log cfu/g	81	7.36	1.26	3.95	9.39
Yeast, log cfu/g	95	4.24	2.06	0.00	7.68
Mold, log cfu/g	73	1.81	1.53	0.00	4.99
Aerobic stability, h	66	81.0	52.6	10.7	240

[1]SCP = soluble CP; WSC = water-soluble carbohydrates; LA:AA = lactic-to-acetic acid ratio; NH_3-N = ammonia-nitrogen; LAB = lactic acid bacteria.

[2]Number of treatment means.

statistical variation around variables of interest. Additional sources of heterogeneity may arise from differences in storage structures, packing density, and on-farm practices. Because our main objective was to identify variation in responses associated with ST, future research could further investigate other factors contributing to heterogeneity in the effect sizes evaluated in this study.

Effects of Storage Temperature on Silage Nutritive Value and Fermentation Profile

Effects of ST on nutritive value of silage stored at different temperature ranges are presented in Table 4. Dry matter concentration was greater (P = 0.001) for silage stored at >40°C (32.5% DM) compared with silage stored at ≤10°C (30.5% DM), 11°C to 20°C (30.1% DM), and 21°C to 30°C (31.0% DM). Silage stored at >40°C also had greater WSC (P = 0.001; 7.76% DM) and lower CP (P = 0.001; 8.72% CP) concentrations than other temperature ranges. Soluble CP concentration increased progressively with increasing ST (P = 0.001). Lignin concentration was lower (P = 0.002) for silage stored at ≤10°C (3.06% DM) than other temperature ranges. Conversely, concentrations of NDF (P = 0.23; 46.8% DM, on average), ADF (P = 0.16; 29.8% DM, on average), starch (P = 0.16; 35.4% DM, on average), and ash (P = 0.33; 4.25% DM, on average) did not differ among ST.

Effects of ST on fermentation profile of silage are presented in Table 5. Silage pH decreased gradually with increasing ST (P = 0.001), where pH was greatest at ≤10°C (5.21), intermediate from 11°C to 20°C (4.52), and lowest from 21°C to 30°C (4.11) and 31°C to 40°C (4.19). Silage stored at >40°C (4.22) did not differ from that stored at 11°C to 20°C, 21°C to 30°C, or 31°C to 40°C. Lactic acid concentration was greatest (P = 0.001) for silage stored at 21°C to 30°C (5.00% DM), intermediate for silage stored at 31°C to 40°C (4.04% DM) and 11°C to 20°C (3.88% DM), and lowest at ≤10°C (3.20% DM). Silage stored at >40°C (4.83% DM) did not differ from 11°C to 20°C, 21°C to 30°C, and 31°C to 40°C. Acetic acid concentration was affected by ST (P = 0.03), but no differences across ST treatments were observed after using the sequentially rejective Bonferroni t-test adjustment (P > 0.05; 1.18% DM, on average). Lactic-to-acetic acid ratio was greater (P = 0.001) for silage stored at >40°C (4.52) and 21°C to 30°C (3.95) than for silage stored at 11°C to 20°C and 31°C to 40°C, whereas silage stored at ≤10°C (3.11) did not differ from any other temperature range. Concentration of n-butyric acid was greater (P = 0.01) for silage stored at 21°C to 30°C (0.67% DM) than for silage at ≤10°C (0.27% DM), 11°C to 20°C (0.26% DM), and 31°C to 40°C (0.28% DM). Propionic acid concentration also differed among ST (P = 0.01), with greater values at 21°C to 30°C (0.19% DM) and 31°C to 40°C (0.20% DM) than at ≤10°C (below limit detection). However, silage stored in temperatures varying from 11°C to 20°C (0.15% DM) and >40°C (0.12% DM) did not differ among treatments. No differences

Table 4. The effects of storage temperature on silage nutritive value

Item[1]	Storage temperature, °C ≤10	11–20	21–30	31–40	>40	SEM[2]	I^2,[3] %	*P*-value
DM, % as-fed	30.5[b]	30.1[b]	31.0[b]	30.9[ab]	32.5[a]	0.73	91.4	0.001
WSC, % DM	4.85[b]	2.19[c]	2.54[c]	2.45[c]	7.76[a]	0.41	23.6	0.001
CP, % DM	9.59[ab]	9.79[a]	9.49[b]	9.55[ab]	8.72[c]	0.42	95.5	0.001
SCP, % DM	—	43.3[c]	48.9[b]	53.4[a]	—	2.18	96.4	0.001
NDF, % DM	48.0	46.2	46.8	46.7	46.1	1.36	91.3	0.23
ADF, % DM	29.2	29.9	30.1	29.7	30.3	2.68	36.9	0.16
Starch, % DM	32.6	33.6	37.9	36.3	36.8	2.16	95.9	0.16
Lignin, % DM	3.06[b]	4.99[a]	5.25[a]	5.46[a]	5.68[a]	1.13	0.00	0.002
Ash, % DM	4.42	4.41	4.34	3.82	—	0.20	66.4	0.33

[a–c]Means with different superscripts within the same row denote an effect of storage temperature. Significance was considered at $P \leq 0.05$.

[1]SCP = soluble CP; WSC = water-soluble carbohydrates.

[2]Greatest SEM.

[3]I^2 = degree of heterogeneity among studies included in the meta-analysis.

were observed across ST treatments for ethanol concentration ($P = 0.69$; 0.82% DM, on average). An effect of ST was observed ($P = 0.001$) for NH_3-N concentration, which was greater for silage stored in temperatures varying from 31°C to 40°C compared with silage stored in temperatures ranging from 11°C to 20°C (7.32 vs. 6.12% N, respectively). However, no differences were observed among other treatments (6.65% N, on average). Moreover, silage stored at ≤10°C and 11°C to 20°C had greater ($P = 0.001$) LAB (7.71 and 7.68 log cfu/g, respectively) and yeast counts (4.09 and 4.24 log cfu/g, respectively) than silage stored at higher temperatures. Mold counts had a similar pattern ($P = 0.02$), with greater counts at ≤10°C (2.41 log cfu/g) and 11°C to 20°C (2.16 log cfu/g) than at 21°C to 30°C (1.47 log cfu/g). Conversely, no effects were observed across ST treatments for DM loss ($P = 0.37$; 3.89% DM, on average).

Effects of Storage Temperature on Nutritive Value, Fermentation Profile, and Aerobic Stability of WPCS

Effects of ST on nutritive value of WPCS are presented in Table 6. Dry matter concentration was greater ($P = 0.001$) for WPCS stored at 21°C to 30°C (34.0% DM) and 31°C to 40°C (33.7% DM) than for WPCS stored at ≤10°C (32.9% DM) and 11°C to 20°C (33.0% DM). Water-soluble carbohydrate concentration was greatest ($P = 0.001$) for WPCS stored >40°C (12.4% DM), intermediate at ≤10°C (6.60% DM), and lowest for WPCS stored at 11°C to 20°C (2.53% DM), 21°C to 30°C (3.18% DM),

Table 5. The effects of storage temperature on silage fermentation profile and microbial counts

Item[1]	Storage temperature, °C ≤10	11–20	21–30	31–40	>40	SEM[2]	I^2,[3] %	*P*-value
pH	5.21[a]	4.52[b]	4.11[c]	4.19[c]	4.22[bc]	0.09	1.10	0.001
Lactic acid, % DM	3.20[c]	3.88[b]	5.00[a]	4.04[b]	4.83[ab]	0.29	31.2	0.001
Acetic acid, % DM	1.40	1.16	1.09	0.98	1.28	0.10	3.13	0.03
LA:AA ratio	3.11[ab]	2.96[b]	3.95[a]	2.60[b]	4.52[a]	1.68	10.3	0.001
n-Butyric acid, % DM	0.27[b]	0.26[b]	0.67[a]	0.28[b]	0.31[ab]	0.14	0.00	0.01
Propionic acid, % DM	ND[4]	0.15[ab]	0.19[a]	0.20[a]	0.12[ab]	0.03	0.71	0.003
Ethanol, % DM	0.72	0.89	0.84	0.83	0.83	0.47	21.5	0.69
1,2-Propanediol, % DM	—	—	—	—	—	—	—	—
NH_3-N, % N	6.53[ab]	6.12[b]	7.26[a]	7.32[a]	6.17[ab]	0.77	25.8	0.001
LAB, log cfu/g	7.71[a]	7.68[a]	6.85[b]	5.54[c]	5.81[c]	0.98	2.29	0.001
Yeast, log cfu/g	4.09[a]	4.24[a]	2.87[b]	2.78[b]	0.75[c]	1.46	3.76	0.001
Mold, log cfu/g	2.41[a]	2.16[a]	1.47[b]	1.72[ab]	1.44[ab]	0.50	10.5	0.02
DM loss, % DM	2.92	3.63	4.74	4.44	3.72	0.57	0.00	0.37

[a–c]Means with different superscripts within the same row denote an effect of storage temperature. Significance was considered at $P \leq 0.05$.

[1]LA:AA = lactic-to-acetic acid ratio; NH_3-N = ammonia-nitrogen; LAB = lactic acid bacteria.

[2]Greatest SEM.

[3]I^2 = degree of heterogeneity among studies included in the meta-analysis.

[4]ND = not detected.

Table 6. The effects of storage temperature on whole-plant corn silage nutritive value

Item[1]	Storage temperature, °C					SEM[2]	I^2,[3] %	*P*-value
	≤10	11–20	21–30	31–40	>40			
DM, % as-fed	32.9[b]	33.0[b]	34.0[a]	33.7[a]	—	1.38	90.2	0.001
WSC, % DM	6.60[b]	2.53[c]	3.18[c]	2.84[c]	12.4[a]	0.88	31.5	0.001
CP, % DM	7.25[b]	7.47[a]	7.13[b]	7.04[b]	6.55[b]	0.34	52.3	0.001
SCP, % DM	—	39.6[c]	45.1[b]	58.9[a]	—	3.07	99.2	0.001
NDF, % DM	41.7	41.1	41.9	39.4	39.9	1.48	71.0	0.07
ADF, % DM	21.5	20.9	21.6	20.4	21.0	0.82	40.6	0.28
Starch, % DM	33.4	34.4	38.7	37.2	—	1.94	96.1	0.09
Lignin, % DM	—	4.61[b]	5.00[a]	5.20[a]	—	0.10	98.1	0.001
Ash, % DM	4.08	4.06	4.01	3.58	—	0.24	74.2	0.46

[a–c]Means with different superscripts within the same row denote an effect of storage temperature. Significance was considered at $P \leq 0.05$.

[1]SCP = soluble CP; WSC = water-soluble carbohydrates.

[2]Greatest SEM.

[3]I^2 = degree of heterogeneity among studies included in the meta-analysis.

and 31°C to 40°C (2.84% DM). Crude protein concentration was greatest ($P = 0.001$) for WPCS stored at 11°C to 20°C compared with all other temperature ranges. Conversely, SCP concentration gradually increased with increasing ST ($P = 0.001$). Lignin concentration was lower ($P = 0.001$) for WPCS stored at 11°C to 20°C (4.61% DM) than at 21°C to 30°C (5.00% DM) and 31°C to 40°C (5.20% DM). No differences across ST treatments were observed for NDF ($P = 0.07$; 40.8% DM, on average), ADF ($P = 0.28$; 21.1% DM, on average), starch ($P = 0.09$; 35.9% DM, on average), and ash ($P = 0.46$; 3.93% DM, on average) concentrations.

Effects of ST on fermentation profile of WPCS are presented in Table 7. Silage pH was greatest ($P = 0.001$) at ≤10°C (5.64), intermediate at 11°C to 20°C (4.56), and lowest at 21°C to 30°C (3.95). Silage stored at 31°C to 40°C (4.07) and >40°C (3.92) did not differ from either 11°C to 20°C or 21°C to 30°C. Lactic acid concentration was affected by ST ($P = 0.001$) and was greatest for WPCS stored at 21°C to 30°C compared with other temperature ranges. No differences across ST treatments were observed for acetic acid concentration ($P = 0.89$; 1.57% DM, on average). Lactic-to-acetic acid ratio was greater ($P = 0.01$) for WPCS stored at 21°C to 30°C (4.69) than at 11°C to 20°C (3.56) and 31°C to 40°C (2.18), whereas WPCS stored at ≤10°C (3.67) and >40°C (3.31) did not differ from other temperature ranges. An effect of ST was observed ($P = 0.01$) for propionic acid concentration, which was greater for WPCS stored at 21°C to 30°C (0.24% DM, on average) than ≤10°C (below limit detection) and >30°C (0.08% DM, on average). However, WPCS stored in temperatures varying from 11°C to 20°C did not differ from other treatments. Ethanol concentration was greater ($P = 0.003$) at 11°C to 20°C (1.16% DM) than at 31°C to 40°C (0.83% DM), whereas WPCS stored at ≤10°C (1.04% DM) and 21°C to 30°C (0.83% DM) did not differ from other temperature ranges. For NH_3-N concentration, the greatest values were observed ($P = 0.001$) for WPCS stored at 21°C to 30°C (5.02% N), 31°C to 40°C (6.25% N), and >40°C (5.15% N), intermediate values were observed at 11°C to 20°C (4.13% N), and the lowest concentrations occurred at ≤10°C (2.67% N). Concentration of 1,2-propanediol differed across ST ($P = 0.001$), being detected only in WPCS stored at 11 to 40°C and therefore greater than at ≤10°C (0.00% DM). Lactic acid bacteria counts were lower ($P = 0.001$) for WPCS stored at 31°C to 40°C compared with other temperature ranges. Yeast counts were greater ($P = 0.02$) for WPCS stored at ≤10°C (5.37 log cfu/g) than when stored at 21°C to 30°C (3.60 log cfu/g) and 31°C to 40°C (3.94 log cfu/g) temperature ranges. Mold count was not affected ($P = 0.35$) by ST. Moreover, both DM loss ($P = 0.001$) and aerobic stability ($P = 0.01$) increased with increasing ST. The increment in DM loss was even more pronounced when ST increased over 40°C.

DISCUSSION

Minimal effects of ST were observed on nutritive value regardless of forage type. Greater DM concentration with increasing ST was previously described in literature. For example, Bai et al. (2022b) reported that DM concentration was greater for silage stored at 30 than 20°C. Similarly, Adesogan (2006) found that storage at high temperatures (40 vs. 20°C) slightly increased DM concentration by 2 percentage units. This was likely due to a drying effect and subsequent evaporation of intracellular water. This commonly occurs in the superficial areas of poorly compacted and sealed commercial silos (McDonald et al., 1991). In addition, greater WSC concentrations in the lowest and highest ST treatments (≤10°C and >40°C, respectively) indicate curtailed fermentation, which is

Table 7. The effects of storage temperature on whole-plant corn silage fermentation profile, microbial counts, and aerobic stability

Item[1]	Storage temperature, °C					SEM[2]	I^2,[3] %	P-value
	≤10	11–20	21–30	31–40	>40			
pH	5.64[a]	4.56[b]	3.95[c]	4.07[bc]	3.92[bc]	0.18	0.62	0.001
Lactic acid, % DM	3.04[b]	4.26[b]	5.63[a]	3.68[b]	3.34[b]	0.48	16.5	0.001
Acetic acid, % DM	1.84	1.60	1.56	1.59	1.27	0.30	6.78	0.89
LA:AA ratio	3.67[ab]	3.56[b]	4.69[a]	2.18[b]	3.31[ab]	0.51	10.8	0.001
n-Butyric acid, % DM	ND[4]	ND	ND	ND	ND	—	—	—
Propionic acid, % DM	ND	0.21[ab]	0.24[a]	0.08[b]	—	0.05	0.28	0.01
Ethanol, % DM	1.04[ab]	1.16[a]	0.96[ab]	0.83[b]	—	0.08	40.1	0.01
1,2-Propanediol, % DM	0.00[b]	0.06[a]	0.07[a]	0.07[a]	—	0.02	0.02	0.001
NH_3-N, % N	2.67[c]	4.13[b]	5.02[a]	6.25[a]	5.15[a]	0.49	78.6	0.001
LAB, log cfu/g	7.53[a]	7.91[a]	7.96[a]	5.57[b]	—	0.18	3.69	0.001
Yeast, log cfu/g	5.37[a]	4.54[ab]	3.60[b]	3.94[b]	—	0.36	8.50	0.02
Mold, log cfu/g	1.32	1.74	2.13	2.38	—	0.39	5.53	0.35
DM loss, % DM	1.71[c]	1.97[c]	4.22[b]	4.59[b]	7.58[a]	0.37	0.00	0.001
Aerobic stability, h	16[b]	52[ab]	108[a]	77[a]	—	27.5	0.00	0.01

[a–c]Means with different superscripts within the same row denote an effect of storage temperature. Significance was considered at $P \leq 0.05$.

[1]LA:AA = lactic-to-acetic acid ratio; NH_3-N = ammonia-nitrogen; LAB = lactic acid bacteria.

[2]Greatest SEM.

[3]I^2 = degree of heterogeneity among studies included in the meta-analysis.

[4]ND = not detected.

possibly due to a reduced number of LAB (Weinberg et al., 1998, 2001) and enhanced proteolysis (Weinberg et al., 2001); thus, making the fermentation less homolactic (McDonald et al., 1966) under temperature extremes. In addition, high ST can reduce substrate availability by denaturing enzymes, which in turn can impair LAB growth (Tao et al., 2012). Reductions in CP concentration are typically associated with increases in both SCP and NH_3-N concentrations, which are products of protein degradation. Proteolysis is triggered by plant and microbial proteases, whereas deamination occurs mainly through microbial metabolism (Grum et al., 1991). The greater lignin concentration with increasing ST agrees with literature (Van Soest, 1965; Garcia et al., 1989), which has been reported in heat-damaged forages. Such responses might be associated with mild nonenzymatic browning or Maillard-type condensation reactions occurring under prolonged exposure to elevated ST, thereby triggering the formation of lignin-like complexes (Sajib and Undeland, 2020). These compounds can bind to carbohydrates and proteins, hence reducing their availability for microbial fermentation.

Furthermore, ST influences silage fermentation (McDonald et al., 1966) through regulation of microbial growth and enzymatic activities (Weinberg et al., 2001; Muck et al., 2003). In a study conducted by Bai et al. (2022a) evaluating the effects of ST on fermentation profile and bacterial community of sorghum-sudangrass silage, changes in operational taxonomic units and diversity indexes were observed when silage was stored at 2 temperatures (10°C vs. 25°C). In our meta-analysis, fermentation patterns (i.e., pH and lactic acid concentration) were also changed due to ST. Overall, pH was lower at greater ST ranges in the current study. Similarly, Ali et al. (2015) observed that pH of corn silage stored at 5°C was greater than 12°C or 18°C. Zhou et al. (2019) also found greater pH values of corn silage stored at 10°C than 20°C. This suggests that low temperatures impair LAB metabolism, lowering lactic acid formation and maintaining higher pH values. In fact, lactic acid production was greatest for silage stored at 21°C to 30°C. Likewise, Bai et al. (2022a) observed greater lactic acid production for sorghum-sudangrass silage stored at 25°C than 10°C. The ST varying from 21°C to 30°C is more appropriate for LAB activity during the fermentation process in silage, as reflected by the higher lactic acid concentration and lower pH. For example, Weinberg and Muck (1996) noted that among shortcomings of LAB inoculants, low temperatures can impair microbial inoculants from dominating silage fermentation. Similarly, Liu et al. (2014) found that ST affected fermentation profile of both sweet corn stalk and whole-plant oat silage. In addition, effects of ST on lactic-to-acetic acid ratio were related to slightly greater lactic acid production for silage stored at 21°C to 30°C, as acetic acid production was similar across treatments.

The concentration of n-butyric acid was greatest in silage stored at 21°C to 30°C. This was somewhat unexpected due to an adequate fermentation profile observed within this ST treatment. It is well accepted in the literature that well-fermented silage is virtually free of n-butyric acid (Kung et al., 2018), which agrees with the results found for WPCS in the current study (below limit detection). The production of n-butyric acid indicates metabolic activity from clostridial organisms, which causes nutrient losses (Pahlow et al., 2003). Some clostridia exhibit saccharolytic activity, fermenting sugars to n-butyric acid; others can convert lactic to n-butyric acid, whereas certain strains are highly proteolytic (Muck, 1988). These fermentation pathways are typically associated with higher pH and increased concentrations of acetic acid, NH_3-N, and SCP in silage. Overall, these changes were not observed in this meta-analysis. Disagreement of expected trends among fermentation products, such as an increase of both n-butyric acid and lactic acid at 21°C to 30°C, was likely a result of combining different silage crops in the main dataset. Meanwhile, no n-butyric acid was observed in the WPCS subset.

Propionic acid concentration below limit detection for silage stored at ≤10°C indicates a downregulation of carbohydrate metabolism, which is consistent with the higher pH values. Carbohydrate metabolism is an important metabolism pathway in the silage process whereby LAB converts WSC into organic acids (Novik and Savich, 2019). Ethanol production differed between WPCS stored at 11°C to 20°C and those stored at 31°C to 40°C. Differences in pH combined with greater ethanol production in WPCS may indicate greater numbers of undesirable microorganisms (Kleinschmit and Kung, 2006), which can cause spoilage in the silage material when exposed to air, as some lactate-assimilating yeasts can grow under these conditions (Pahlow et al., 2003). Ethanol production in silage reflects both the metabolic activity of these organisms (enterobacteria and yeasts) and the balance between heterofermentative and homofermentative LAB. In the present study, lower ST appeared to favor certain yeasts and heterofermentative bacteria while also prolonging an early high-pH environment that supports enterobacteria growth (Pahlow et al., 2003). The activity of these microorganisms likely contributed to the greater ethanol concentration observed at 11°C to 20°C. In contrast, greater temperatures typically support homofermentative LAB, which direct carbon primarily toward lactic acid rather than ethanol. Under these conditions, declining pH and intensified competition from LAB reduce the growth and viability of enterobacteria and yeasts (Heron et al., 1993; Pahlow et al., 2003), causing lower ethanol production. Recent metagenomic work demonstrates this temperature-driven restructuring of microbial communities and fermentation gene profiles (Bai et al., 2022a,b; Hu et al., 2024; Zhu et al., 2025). Thus, the decrease in ethanol concentration reflects both a reduction in yeast viability and a functional shift in the bacterial community toward more efficient lactic acid fermentation pathways.

Several microorganisms can produce 1,2-propanediol, such as LAB, clostridia, and yeast species (Suzuki and Onishi, 1968; Sánchez et al., 1987). Epiphytic or inoculated populations of Lactobacillus *buchneri* can yield 1,2-propanediol in silage by converting 1 mol of lactate into half a mole each of acetate, 1,2-propanediol, and carbon dioxide, without an electron acceptor (Oude Elferink et al., 2001). Furthermore, detection of 1,2-propanediol can also suggest that the naturally occurring microorganism, Lactobacillus *diolivorans*, may be present in the silage, which metabolizes this compound, producing 1-propanol and propionic acid (Krooneman et al., 2002). Reduced 1,2-propanediol concentration associated with lack of effects on acetic acid concentration in WPCS indicates decreased LAB activity under low temperatures. Similarly, Zhou et al. (2016) reported that low temperatures (5°C and 10°C) impaired silage fermentation by reducing the growth rate and enzymatic activity of LAB population. In addition, NH_3-N concentration increased for silage stored at ≥20°C, regardless of forage type. Higher ST often promotes protein degradation in silage by stimulating the activity of plant and microbial proteases. Typically, undesirable microorganisms (i.e., *Clostridium* spp.) are influenced by ST, especially at the beginning of silage fermentation. *Clostridium* spp. thrives at high temperatures (>30°C), with optimal growth occurring at 37°C (McDonald et al., 1991). Therefore, in silage stored at high ST, these undesirable microorganisms might be abundant and cause changes in fermentation patterns (McDonald et al., 1991).

Furthermore, LAB counts were reduced in silage stored at high temperatures. McDonald et al. (1991) previously reported that LAB counts may be inhibited by low pH, which aligns with the lower pH values observed under these storage conditions. However, LAB showed lower activity in silage stored at low temperatures (Jungbluth et al., 2017), which may be reflected by lower concentrations of organic acids. Cao et al. (2011) evaluated the effects of low temperatures on the fermentation of TMR silage and found that the LAB activity was lower for silage when temperature reached a minimum of −2°C. Yeast and mold counts in silage decreased with increasing ST, except for mold counts in WPCS. Previous studies found similar responses when evaluating the effects of different ST on dynamic changes in silage fermentation (Ren et al., 2019; Wang et al., 2019). Similarly, Ferrero et al. (2021) reported that both yeast and mold can survive longer at low temperatures. These same authors suggested that lower ST enhances yeast and mold survival by slowing metabolic activity, thereby optimizing the energy-dependent efflux of protons released by organic acids from the cytoplasm. Greater WSC concentration at lower ST (i.e., ≤10°C) compared with other ST treatments may support this survival strategy by providing an energy source for basal metabolism in yeasts and molds. In addition, greater numbers of yeasts are commonly associated with high concentrations of ethanol, and their numbers are often inversely related to the aerobic stability of silage (Kung et al., 1998). In this current study, aerobic stability was lowest for WPCS stored at temperatures ≤10°C. Aerobic deterioration of silage involves complex microbiological processes, which can be influenced by pH, activity of aerobic bacteria and fungi, and concentrations of lactic acid and residual WSC (Muck et al., 1991). In this meta-analysis, WPCS stored at low temperatures (≤10°C) had higher pH values associated with reduced lactic acid concentration and increased yeast counts. These findings indicate poor silage fermentation, which reduces silage nutritive value (Weinberg et al., 2001). Furthermore, DM loss gradually increased with ST in WPCS. Similarly, Rees (1982) observed that a rise of 10°C above the critical temperature (~20°C) can cause DM loss up to 1.7%. Alternatively, Zhou et al. (2019) found that increasing ST can stimulate the growth of *L. buchneri,* which may become dominant under such silage fermentation conditions. This could explain improved aerobic stability and greater DM loss for WPCS stored at higher temperatures. Typically, *L. buchneri* converts lactic acid into acetic acid, 1,2-propanediol, and CO_2 (Oude Elferink et al., 2001). Acetic acid is known to have good antifungal properties and inhibits the growth of yeasts in silage, improving aerobic stability (Woolford, 1975; Moon, 1983). Moreover, heterofermentative bacteria (i.e., *L. buchneri*) have less efficient pathways that can cause slightly greater DM loss during anaerobic storage phase (Pahlow et al., 2003).

These findings suggest that low temperatures can restrict silage fermentation by prolonging the high-pH phase and favoring undesirable microorganisms such as enterobacteria and certain yeasts. Such conditions are particularly challenging in colder regions such as North America, Canada, and Northern Europe (Bernardes et al., 2018), where silage often shows higher pH values associated with delays in the pH decline (Zhou et al., 2016), reduced organic acid concentration (Ali et al., 2015), and higher yeast counts (Ferrero et al., 2021). These fermentation limitations reduce nutritive value and aerobic stability. In contrast, high temperatures stimulate the growth of spoilage organisms, and minimal effects on fermentation profile are unclear in this study. For example, Koc et al. (2009), evaluating the effects of temperature on fermentation profile of WPCS, observed greater yeast and mold counts and CO_2 production at a higher (30°C–37°C) compared with a lower temperature (20°C). Across both thermal extremes, producers may improve fermentation outcomes by ensuring rapid silo filling, maximizing packing density to limit oxygen, and avoiding premature feed-out. In cold climates, additional consideration should be given to extending storage duration and using inoculants containing cold-tolerant heterofermentative strains that can initiate and sustain fermentation at low temperatures. Conversely, in warm or tropical regions, emphasis should be placed on rapid sealing and careful face management to prevent excessive heating, and the use of heterofermentative inoculants that enhance aerobic stability (e.g., *Lactobacillus buchneri*) can provide further benefits. Producers should also monitor silage temperature during storage and feed-out, as excessive heating not only promotes nutrient losses but may also indicate oxygen infiltration that warrants correction of management practices. Together, these practices may help mitigate temperature-related challenges and promote more consistent fermentation across diverse environments.

CONCLUSIONS

Higher temperatures have been associated with reduced LAB counts, whereas low temperatures may not necessarily reduce LAB populations but can alter their metabolic activity. Changes in fermentation profiles across these ST treatments suggest a possible shift in dominant bacterial species in silage and the need for bacterial inoculants that could modulate fermentation under these conditions. Therefore, the temperature range of 21°C to 30°C may be more suitable for silage storage, thereby enhancing nutrient preservation. In cold climates, producers should allow longer storage times, ensure adequate packing, and consider inoculants designed for improved performance under low-temperature conditions. In contrast, in warm or tropical regions, rapid silo sealing and careful face management are important to prevent excessive heating and nutrient losses during feed-out.

NOTES

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Nonstandard abbreviations used: LA:AA = lactic-to-acetic acid ratio; LAB = lactic acid bacteria; PRISMA = preferred reporting items for systematic reviews and meta-analysis; RMSE = root mean square error; SCP = soluble CP; ST = storage temperature; WPCS = whole-plant corn silage; WSC = water-soluble carbohydrates.

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ORCIDS

G. F. L. Cruz, https://orcid.org/0000-0001-8977-1765
M. R. Pupo, https://orcid.org/0000-0003-1209-6145
J. L. P. Daniel, https://orcid.org/0000-0003-0569-899X
L. F. Ferraretto https://orcid.org/0000-0003-4711-8332